

design representation and analysis, the core is based on a powerful, persistent and scalable design database, ZDB. Unlike other open source tools, ZamiaCAD has its own built-on object database based on Java serialisation technology, helping design engineers with its fully elaborated design model and the useful static analysis support tasks, such as global signal reference search and FSM recognition. Additionally, the core also includes some waveform file import modules and a built-in simulator for ad-hoc validations.

The simulator applications, along with the Eclipse GUI, are built on top of the IG and the language-dependent abstract syntax tree structures in the frontend. This supports graphical viewers and editors with an automatic model builder for their ease of modelling and design by utilising all the functionality of the other blocks.

Another significant feature of ZamiaCAD comes in the form of a very high-speed VHDL indexer that extracts the information about which design units are declared in which files. The actual VHDL parser can then be used whenever required on local source files, or those in your eclipse workspace, and the files defining design units that are actually required during elaboration.

Features for your ease of use

ZamiaCAD is a tool for advanced hardware design, analysis and research. It can be beneficial for developers writing EDA tools, end-users using EDA tools coming with the platform and also the researchers prototyping research algorithms. ZamiaCAD supports many

Special advantages

ZamiaCAD adheres to its high performance, scalability and usability goals by its various design principles. It features many advantages over its competitors.

Non-invasive designing. One of the major design principles of ZamiaCAD is non-invasiveness. Any engineer can start working with ZamiaCAD using pre-existing projects with minimum configuration effort as HDL being one and only data format. It also ensures that no other member of the team is forced to work on ZamiaCAD if one is using it.

Scalability and high performance. Scalability and high performance targets are ensured by the IG structures used for intermediate design representation along with a powerful, persistent and scalable design database.

Supports large designs. ZamiaCAD can also transcend memory limitations in case of particularly large designs as it is empowered by efficient Java serialisation technology.

Robustness. Another advantage of Java is that, JUnit tests exist for many parts of the Java framework, which accounts for robustness of the tool.

features that can be of great help while working with the tool, such as:

Full design-entry support. ZamiaCAD supports syntax highlighting and auto-completion functions with the help of easily-extensible VHDL templates and also the outlines. These design-entry-support functions account for an incremental model builder approach and easy error marking.

Easy design navigation. ZamiaCAD allows you to navigate easily through your designs and trace your signals and references. You can open your design units and signal declarations using instantiation graph (IG) and hierarchical path.

Handy simulator. The built-in reference simulator, along with waveform viewer and waveform file import functionality, accounts for handy simulations in ZamiaCAD. You can also have source back-annotation of simulated values.

Complete design analysis. Design analysis can be seen in parallel with navigation. You can have reference search for instantiated modules and signals across hierarchy levels in the design. It covers almost all the features of

any Eclipse framework with additional functionalities like declaration search and file caching to mitigate latencies.

A powerful non-invasive tool

ZamiaCAD is a powerful non-invasive tool made for design and verification engineers. It is well suited for the automation of manual design and verification tasks in the design areas like entry, analysis, integration of designs and simulations. This can help in increasing register transfer level design and verification engineers' productivity.

With the complete VHDL standard supported as hardware description language and options to extend it by adding other frontends, such as Verilog, ZamiaCAD qualifies well for being a good option for advanced hardware designs.

With its structure and design principles, ZamiaCAD tool supports the state-of-the-art and future large-scale industrial hardware designs. It is equipped with powerful debugging functionality and provides the research community with an open source robust research platform.

Support

ZamiaCAD is a free GPL licensed software tool that can be downloaded at <http://zamiacad.sourceforge.net/web/>. Also, you can find developer's documentation and some quick-start tutorials available on the website for you to get started with ZamiaCAD and know how it looks and feels like. ●

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Installing ZamiaCAD

Installation file can be found in EFY Plus DVD that came with this magazine. The DVD includes ZamiaCAD bundled as a product that includes Eclipse with embedded ZamiaCAD plugin.

Supported OS: Windows, Linux, MacOS

HDD space required: About 100MB

Note. If you already have Eclipse running and you want to integrate ZamiaCAD into your Eclipse, you can install ZamiaCAD plug-in using the following steps: Open menu Help→Install New Software... and enter location for your ZamiaCAD files, for both Name and Location in the Add window. Alternatively, you can also enter the website url of the latest version of ZamiaCAD download location. After completing installation, open Window→Open Perspective→Others→ZamiaCAD.